Dawid Sac

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I am a computer science student. I am passionate about informatics and electronics, and my goal is to bridge the gap between both worlds by pursuing a career in **Embedded Systems** and **Firmware Engineering**. I continuously push myself through complex personal projects, constantly exploring new technologies and using my mistakes as stepping stones to improve. Explore my work and learn more about my background at **dawid-sac.dev**.

Education

Haute École Léonard de Vinci Sep. 2024 – Now
*B.Sc. in Applied **Computer Science*** *Brussels, Woluwe-Saint-Lambert*

- **Courses**: Linux Systems, C Programming, Data Structures & Algorithms, Web Development, DevOps

INRACI (Institut de Radioélectricité et Cinématographie) Sep. 2021 – Jun. 2024
*Technical Qualification Degree in **Electronics** (CESS)* *Brussels, Forest*

- **Courses**: Analog & Digital Electronics, Arduino Development, Schematic & PCB Design, Prototyping, 3D CAD
- **Academic distinction**: **Jury Prize** for Best Final-Year Project.

Projects

SmartFrame | *Connected LED Matrix Display* Sep. 2026

- **Tech Stack:** *C, ESP-IDF, FreeRTOS, ESP32, HTTP Server, Spotify & OpenWeather APIs*
- Engineered a multithreaded **ESP32** application in **FreeRTOS** driving a 64-LED matrix to display real-time Spotify tracks, weather data, and custom animations; hosted an **embedded web server** for live device configuration and implemented **custom memory allocators** to optimize RAM.

Auto-Mate | *Automated Chessboard* Jun. 2025

- **Tech Stack:** *C++, Arduino, KiCad, PCB Design, Hardware Debugging*
- Co-developed an automated chessboard from scratch—designing the electrical schematic and **custom PCB**, soldering and debugging hardware, and programming the piece-movement engine in **C++**. Awarded the **Jury Prize** for Best Final-Year Project.

MBPlayer | *Portable Digital Audio Player* In Progress

- **Tech Stack:** *C, STM32, CubeIDE, CubeMX, KiCad*
- Developing a custom hardware digital audio player powered by an **STM32** microcontroller; currently designing the multi-layer **PCB** and writing low-level firmware for audio decoding and peripheral control.

Portfolio dawid-sac.dev | *3D Event-Driven Web Platform* In Progress

- **Tech Stack:** *TypeScript, Three.js, React, HTML/CSS*
- Developing an **engine-like 3D architecture** and **event-handling system** with **custom shaders**, allowing users to interactively explore showcased projects and detailed engineering breakdowns.

Technical Skills

Programming Languages: C, C++, Java, C#, TypeScript, JavaScript, Python, SQL, HTML/CSS
Embedded & Firmware: ESP-IDF, FreeRTOS, STM32, Arduino, I2C, SPI, SDMMC,UART, PWM
Electronics & Lab: Oscilloscope, Function Generator, Multimeter, Through-Hole & SMD Soldering, Hardware Debugging
CAD & Prototyping: KiCad (Schematic & PCB Design), 3D Modeling (CAD), 3D Printing, Technical Documentation
Frameworks & Libraries: Node.js, React, Express.js, Three.js, Flutter, RabbitMQ
Tools & Systems: Git, GitHub, Docker, CMake, Make, Linux (Debian), Apache, Unity
Spoken Languages: French (Fluent), Polish (Native), English (B2+)

Experience

Private Computer Science Tutor Sep. 2025 – Jan. 2026
One-on-one coaching for a 1st-year B.Sc. Computer Science student *Brussels*

Private Mathematics & Physics Tutor May 2024 – Jun. 2025
Academic coaching for a 3rd-year secondary school student *Brussels*

Dishwasher and Waiter May 2025 – Jul. 2026
Walkin'Thai *Brussels, Ixelles*

Convenience Store Cashier Aug. 2024 – Mar. 2025
Chez Pawel *Brussels, Uccle*